Exploring Heavy Tails
Pareto and Generalized Pareto Distributions

September 25, 2019

This vignette is designed to give a short overview about Pareto Distributions and Generalized Pareto Distributions (GPD). We will work with the `SPC.we` data of our **quantmod** vignette. Therefore we have to reproduce the `SPC.we` data in exactly the same way as described the **quantmod** vignette.

In financial data analysis stock indices as the S&P 500 index are typically analyzed by using the *returns* of the index. We use the log-returns

```
> WSPLRet <- diff(log(SPC.we))
```

We start to analyze these by plotting a histogram

```
> hist(WSPLRet)
```

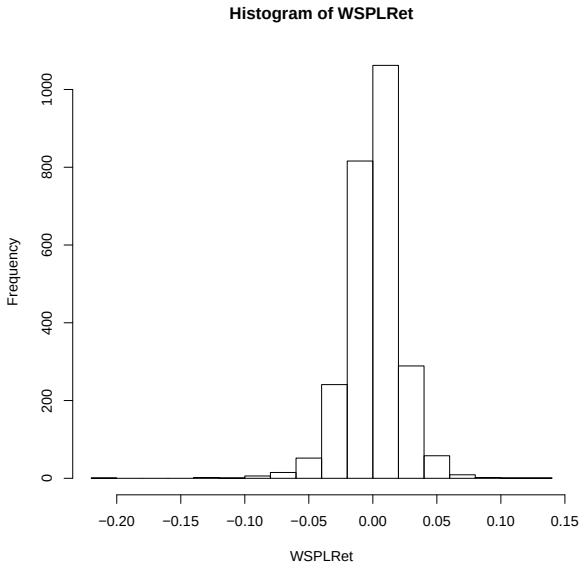


Figure 1: Histogram of the log-returns of the S&P 500 from 1960-01-04 to 2009-01-01.

This histogram shows a unimodal distribution of values with the peak around 0, which nourishes the hypothesis that the log-returns are normally distributed. A very intuitive method to test this is the Q-Q plot.

The slope of the (linear regression) line and its intercept determine the parameters of the corresponding Gaussian distribution. If the points are close to this line the empirical distribution of the sample can

```
> qqnorm(WSPLRet)
> qqline(WSPLRet)
```

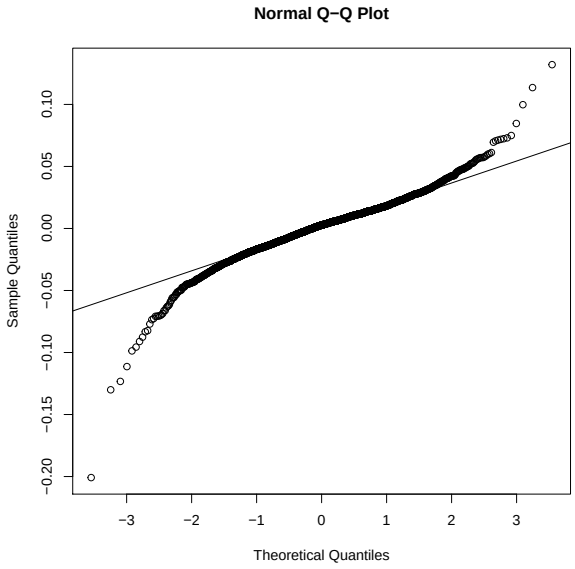


Figure 2: Q-Q plot of `WSPLRet` values.

very well be approximated by a normal distribution. Figure 2 shows that log-returns of the weekly S&P 500 index have heavy tails on both sides and are therefore not modeled well by a normal distribution. The tails of the normal distribution are too thin to produce enough extreme events to match those in the sample.

However, other families of distributions, like Pareto distributions can be used. One way to identify classes of distributions which produce wild events is to show that the density of the considered distribution decays polynomially and then to estimate the degree of such a polynomial decay (Note that for the normal distribution, decay is exponential). Such distributions are called *generalized Pareto distributions* (GPD).

In the following we give a short explanation of Pareto Distributions and GPDs, before we study the problem of estimating the tails of or S&P 500 returns.

1 Pareto distribution

The Pareto distribution (e.g., `https://en.wikipedia.org/wiki/Pareto_distribution`) is commonly used for quantities that are distributed with very long right tails. It is named after the Italian economist Vilfredo Pareto, who originally used this distribution to describe the allocation of wealth among individuals since it seemed to show rather well the way that a larger portion of the wealth of any society is owned by a smaller percentage of the people in that society.

A random variable X has a Pareto distribution with *scale* parameter $K > 0$ and *shape* parameter $\alpha > 0$ iff its cumulative distribution function is given by

$$F(x) = \begin{cases} 1 - (K/x)^{\alpha}, & x \geq K \\ 0, & x < K. \end{cases}$$

(If a family of probability distributions with parameter s and other parameters θ is such that the cumulative distribution functions satisfy $F_{s,\theta}(x) = F_{1,\theta}(x/s)$, then s is a scale parameter. In the above, note that for $x \geq K$, $F_{K,\alpha}(x) = 1 - (x/K)^{-\alpha} = F_{1,\alpha}(x/K)$.)

Hence, K is the minimum possible value of X. The density of X is then given by

$$f(x) = \begin{cases} \alpha K^{\alpha}/x^{\alpha+1}, & x \geq K \\ 0, & x < K. \end{cases}$$

For a shape parameter $\alpha > 1$ the expected value is given by

$$\mathbb{E}(X) = \frac{\alpha K}{\alpha - 1},$$

otherwise ($\alpha \leq 1$) the expected value is infinite.

How the probability distribution of the Pareto distribution changes when one varies the shape parameter is illustrated in the following example where we make use of function `dpareto()` included in package **mistr**:

```
> library("mistr")
> x <- seq(0.1, 10, length = 1000)
> plot(x, dpareto(x, scale = 1, shape=1),
+      type = "l", xlab = "x", ylab = "dpareto(x)",
+      main = "Pareto Probability Density")
> lines(x, dpareto(x, scale = 1, shape=.5), col = "red")
> lines(x, dpareto(x, scale = 1, shape= .2), col = "blue")
> legend("topright", legend = c(1, 0.5, 0.2), col = c(1, 2, 4), lty = 1)
```

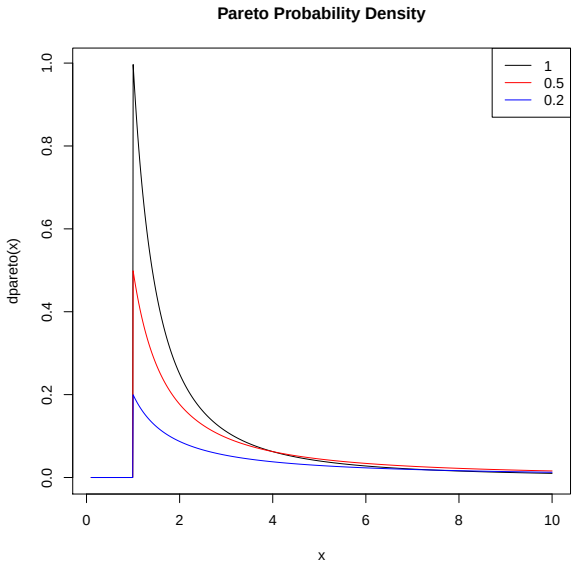


Figure 3: Pareto probability density for shape parameters equal to 1, 0.5, and 0.2.

2 Generalized Pareto Distribution

In comparison to the Pareto Distributions, the Generalized Pareto Distribution (GPD, e.g., `https://en.wikipedia.org/wiki/Generalized_Pareto_distribution` has three three parameters; one *location parameter* μ and two parameters for *scale* and *shape*, σ and ξ. The cumulative distribution function of the GPD is given by:

$$\mathbb{P}(X \leq x) = \begin{cases} 1 - \left(1 + \xi \frac{x-\mu}{\sigma}\right)^{-1/\xi}, & \xi \neq 0 \\ 1 - \exp\left(-\frac{x-\mu}{\sigma}\right), & \xi = 0, \end{cases}$$

for $x \geq \mu$ when $\xi \geq 0$, and $\mu \leq x \leq \mu - \sigma/\xi$ when $\xi < 0$, where μ and ξ are arbitrary real numbers and $\sigma > 0$.

(Note that the distribution function must take values in $[0, 1]$. For $\xi > 0$, this needs $1+\xi(x-\mu)/\sigma \geq 1$, which is equivalent to $x \geq \mu$. For $\xi < 0$, this needs $0 \leq 1 + \xi(x - \mu)/\sigma \leq 1$, which is equivalent to $\mu \leq x \leq \mu - \sigma/\xi$.)

For a $\xi < 1$, the mean of a GPD is given by

$$\mathbb{E}(X) = \mu + \frac{\sigma}{1 - \xi}.$$

The GPD is generalized in the sense that it contains a number of special cases: When $\xi > 0$ and $\mu = 0$, the distribution function is that of an ordinary Pareto Distribution with $\alpha = 1/\xi$ and $K = \sigma/\xi$.

If we are interested in generating generalized Pareto random variables we can apply the following formula:

$$X = \mu + \frac{\sigma(U^{-\xi} - 1)}{\xi} \sim GPD(\mu, \sigma, \xi)$$

for a uniformly distributed variable $U \sim \text{unif}(0, 1)$.

Back to the S&P 500: Like the exponential distribution, the Generalized Pareto distribution is often used to model the tails of another distribution. Now we will use the GPD in order to understand the tails of the log-returns of the S&P 500 index as described in the **quantmod** vignette.

For this purpose, we assume a three components composite model (mixture model with truncated components onto a disjoint support). The first and third component will be used to model the extreme cases, i.e., tails, and the second component will try to catch the center of the empirical distribution. Density function of such a distribution can be written as:

$$f(x) = \begin{cases} w_1 \frac{f_1(x)}{F_1(\beta_1)} & \text{if} \quad -\infty < x < \beta_1, \\ w_2 \frac{f_2(x)}{F_2(\beta_2) - F_2(\beta_1)} & \text{if} \quad \beta_1 \leq x < \beta_2, \\ w_3 \frac{f_3(x)}{1 - F_3(\beta_2)} & \text{if} \quad \beta_2 \leq x < \infty, \end{cases}$$

where $f_i(x)$ and $F_i(x)$ are the PDF and CDF of the i-th component, and β_i and w_i are the i-th breakpoint and weight, respectively.

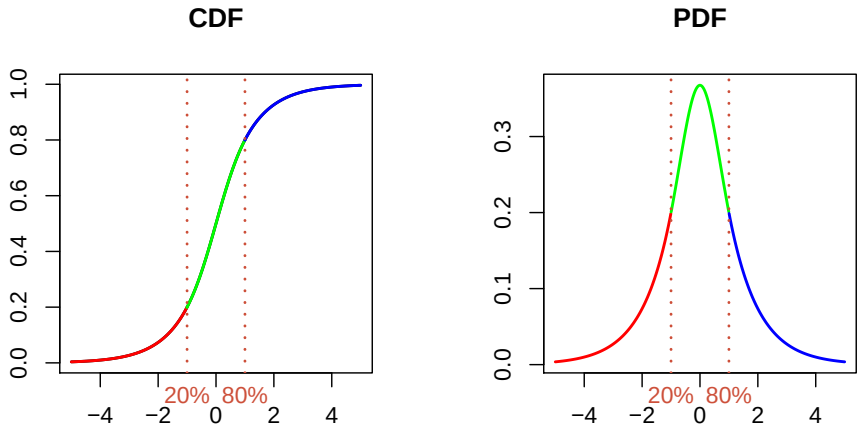
To better understand the model, it might be useful to visualize the distribution and density of such a model. Assume that

- f_1 is a density function of a random variable $-X$, where X follows exponential distribution with rate parameter $\lambda = 1$,
- f_2 is a density function of a Student-t distribution with degrees of freedom equal to 2,

- f_3 is a density function of an exponentially distributed random variable with rate $\lambda = 1$,
- weights are distributed as 20%, 60% and 20% for first, second and third component respectively,
- breakpoints are fixed to be $\beta_1 = -1$, $\beta_2 = 1$.

Then the distribution can be visualized using **mistr** package as:

```
> dist <- compdist(-expdist(1), tdist(2), expdist(1),
+                  weights = c(0.2, 0.6, 0.2),
+                  breakpoints = c(-1, 1))
> plot(dist, xlim1 = c(-5, 5), xlab1 = "", ylab1 = "", xlab2 = "", ylab2 = "")
```



(Note that even though the density function is built like a lego, it still integrates to one and hence is a proper distribution.)

As the Q-Q plot suggests, while the center of the distribution is very well explained using the normal distribution, the tails are heavier and the Pareto family of distributions might be better for those parts. The package **mistr** offers two functions/models for such a problem.

The first offered model is the Pareto-Normal-Pareto (PNP) model. This means that a $-X$ transformation of a Pareto random variable will be used for the left tail, normal distribution for the center and again Pareto for the right tail. From this it follows that the PDF of the model can be written as:

$$f(x) = \begin{cases} w_1 \frac{f_{-P}(x)}{F_{-P}(\beta_1)} & \text{if} \quad -\infty < x < \beta_1, \\ w_2 \frac{f_N(x)}{F_N(\beta_2) - F_N(\beta_1)} & \text{if} \quad \beta_1 \leq x < \beta_2, \\ w_3 \frac{f_P(x)}{1 - F_P(\beta_2)} & \text{if} \quad \beta_2 \leq x < \infty, \end{cases} \tag{1}$$

where $f_P(x) = f_{-P}(-x)$ and $F_P(x)$ are density and distribution function of a Pareto distribution and $F_{-P}(x) = 1 - F_P(-x)$. $f_N(x)$ and $F_N(x)$ are the PDF and CDF of the normal distribution, respectively.

If we follow the properties of the Pareto distribution, the conditional probability distribution of a Pareto-distributed random variable, given the event is greater than or equal to $\gamma > K$, is again a Pareto distribution with parameters γ and α. This means that in (1), the conditional distribution $f_P(x|K, \alpha)/(1 - F_P(\beta_2|K, \alpha)) = f_P(x|\beta_2, \alpha)$ if $\beta_2 > K$. On the other hand, if $\beta_2 < K$ the distribution

cannot be continuous as the support of the Pareto distribution starts at K. The same can be shown for the transformed distribution as

$$\frac{f_{-P}(x)}{F_{-P}(\beta_1)} = \frac{\frac{\alpha K^\alpha}{(-x)^{\alpha+1}}}{1-\left(1-\left(\frac{K}{-\beta_1}\right)^\alpha\right)} = \frac{\frac{\alpha K^\alpha}{(-x)^{\alpha+1}}}{\left(\frac{K}{-\beta_1}\right)^\alpha} = \frac{\alpha\left(-\beta_1\right)^\alpha}{(-x)^{\alpha+1}} = f_{-P}(x|-\beta_1,\alpha) \text{ if } K<-\beta_1.$$

Since we are interested only in the continuous case we can rewrite the PDF (1) as

$$f(x) = \begin{cases} w_1 f_{-P}(x|-\beta_1,\alpha_1) & \text{if } -\infty < x < \beta_1, \\ w_2 \frac{f_N(x|\mu,\sigma)}{F_N(\beta_2|\mu,\sigma)-F_N(\beta_1|\mu,\sigma)} & \text{if } \beta_1 \le x < \beta_2, \\ w_3 f_P(x|\beta_2,\alpha_2) & \text{if } \beta_2 \le x < \infty, \end{cases} \quad \text{where} \quad \begin{array}{c}\beta_1<0<\beta_2,\\ \alpha_1,\alpha_2>0.\end{array}$$

The condition $\beta_1 < 0 < \beta_2$ follows from the fact that the scale parameter has to be positive. Thus, such a model can be fully used only with demeaned data sample or with data with a mean close to zero. This is of course not a problem for stock returns, which are the aim of this vignette. What is more, one can show that the density is continuous if it holds for the shape parameters that

$$\alpha_1 = -\beta_1 \frac{w_2 f_N(\beta_1|\mu,\sigma)}{w_1\left(F_N(\beta_2|\mu,\sigma)-F_N(\beta_1|\mu,\sigma)\right)}, \quad \alpha_2 = \beta_2 \frac{w_2 f_N(\beta_2|\mu,\sigma)}{w_3\left(F_N(\beta_2|\mu,\sigma)-F_N(\beta_1|\mu,\sigma)\right)}.$$

Due to the fact that a composite distribution can be represented as a mixture of truncated distributions that are truncated to disjoint support, the weights for each component can be estimated as the proportion of points that correspond to each of the truncated region. Obviously, this condition ensures that the empirical and estimated CDF match at each of the breakpoints. Thus, conditionally on the fact that the breakpoints are known, the weights can be computed as

$$w_1 = \frac{\sum_{i=1}^n 1_{\{x_i<\beta_1\}}}{n},\ w_2 = \frac{\sum_{i=1}^n 1_{\{\beta_1\le x_i<\beta_2\}}}{n} \text{ and } w_3 = \frac{\sum_{i=1}^n 1_{\{\beta_2\le x_i\}}}{n}, \tag{2}$$

where $1_{\{.\}}$ is the indicator function, and x_i is the i-th data value. These conditions decrease the number of parameters from 11 to 4, and imply the density function of a form $f(x|\beta_1,\beta_2,\mu,\sigma)$.

This model if offered by the **mistr** package using the call `PNP_fit()`. The function `PNP_fit()` takes the data and a named vector of starting values with names break1, break2, mean, and sd and returns a list of class *comp_fit*. Other arguments are passed on to the optimizer.

```
> PNP_model <- PNP_fit(WSPLRet, start = c(break1 = -0.03, break2 = 0.03,
+                                         mean = 0, sd = 0.017))
> PNP_model

Fitted composite Pareto-Normal-Pareto distribution:

Breakpoints: -0.032238 0.028318
Weights: 0.049687 0.882238 0.068075

Parameters:
  scale1   shape1     mean       sd   scale2   shape2
0.032238 2.292886 0.001899 0.017893 0.028318 3.050121

Log-likelihood: 6375.62,  Average log-likelihood: 2.4944
```

If the fitted object is printed, the function will print all the parameters together with the log-likelihood that was achieved by the optimization. In addition, the average log-likelihood is printed, which is just the

log-likelihood divided by the size of the data-set. The user can extract parameters using the call `parameters()`, weights using the call `weights()`, and breakpoints using `breakpoints()`. The `distribution()` call can be used for extracting the distribution with fitted parameters that can be used for evaluation.

Finally, the `plot()` function is offered. The functions plot the Q-Q plot of the fitted distribution and data, and the PDF and CDF plot of the fitted distribution, which overlap with the empirical CDF and PDF of the data-set. The calls contain argument `which`, that extract the proposed plots separately (i.e., `which = "pdf"`). Other arguments are passed on to the the plot calls.

```
> plot(PNP_model)
```

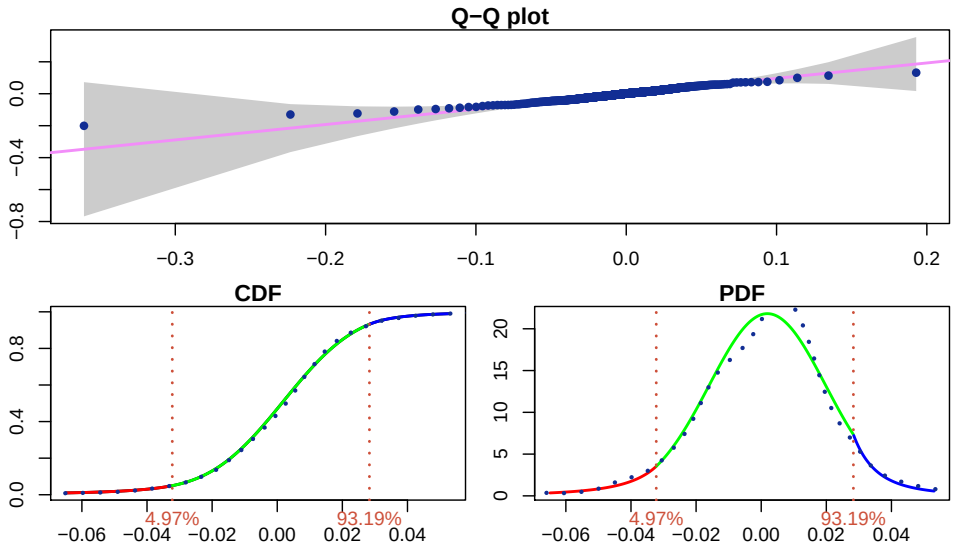


Figure 4: Fitted PNP model.

The second offered model is a similar distribution to the previous one, except we will replace the Pareto distributions by the generalized Pareto distributions (GPD). This means that the PDF of this model can be written as:

$$f(x) = \begin{cases} w_1 \frac{f_{-GPD}(x)}{F_{-GPD}(\beta_1)} & \text{if} \quad -\infty < x < \beta_1, \\ w_2 \frac{f_N(x)}{F_N(\beta_2) - F_N(\beta_1)} & \text{if} \quad \beta_1 \leq x < \beta_2, \\ w_3 \frac{f_{GPD}(x)}{1 - F_{GPD}(\beta_2)} & \text{if} \quad \beta_2 \leq x < \infty. \end{cases}$$

The same way as in the PNP model, the scale parameters can be eliminated by the continuity conditions, weights by the (2) and in addition, under current settings and the continuity conditions the value of the conditional GPD distribution depends on the location parameter only through the conditions $-\beta_1 \geq \theta_1$ and $\beta_2 \geq \theta_2$. This offers to choose without any loss in the model $-\beta_1 = \theta_1$ and $\beta_2 = \theta_2$. Such a PDF is fully characterized by $f(x|\beta_1, \beta_2, \mu, \sigma, \xi_1, \xi_2)$, where the only restriction on the parameters is $-\infty < \beta_1 < \beta_2 < \infty$.

These conditions decrease the number of parameters from 13 to 6. What is more, the function `GNG_fit()` contains the argument `break_fix`, which fixes the breakpoints from the vector of starting values, and hence decreases the number of parameters to 4 if `TRUE` is assigned. In this case, the breakpoints are fixed and weights are computed before the optimization. The function `GNG_fit()` takes the data,

the named vector of starting values with names `break1`, `break2`, `mean`, `sd`, `shape1` and `shape2`, the `break_fix` argument and the argument `midd`, which is by default set to be equal to the mean of the data. The `midd` values are used to split $\mathbb{R}$ into two subintervals and then the first breakpoint is optimized on the left of the midd value and the second breakpoint on the right.

The call returns a list of class *comp_fit*. The results can be then extracted, printed or visualized in the same way as the results of `PNP_fit()`.

```
> GNG_model <- GNG_fit(WSPLRet, start = c(break1 = -0.02, break2 = 0.02, mean = 0,
+                                         sd = 0.016, shape1 = 0.16,  shape2 = 0.11))
> GNG_model

Fitted composite GPD-Normal-GPD distribution:

Breakpoints: -0.021677 0.019926
Weights: 0.109155 0.748435 0.14241

Parameters:
    loc1    scale1    shape1      mean        sd      loc2    scale2    shape2
0.021677 0.013211 0.156440 0.002649 0.017073 0.019926 0.010424 0.123679

Log-likelihood: 6384.591,  Average log-likelihood: 2.4979


> plot(GNG_model)
```

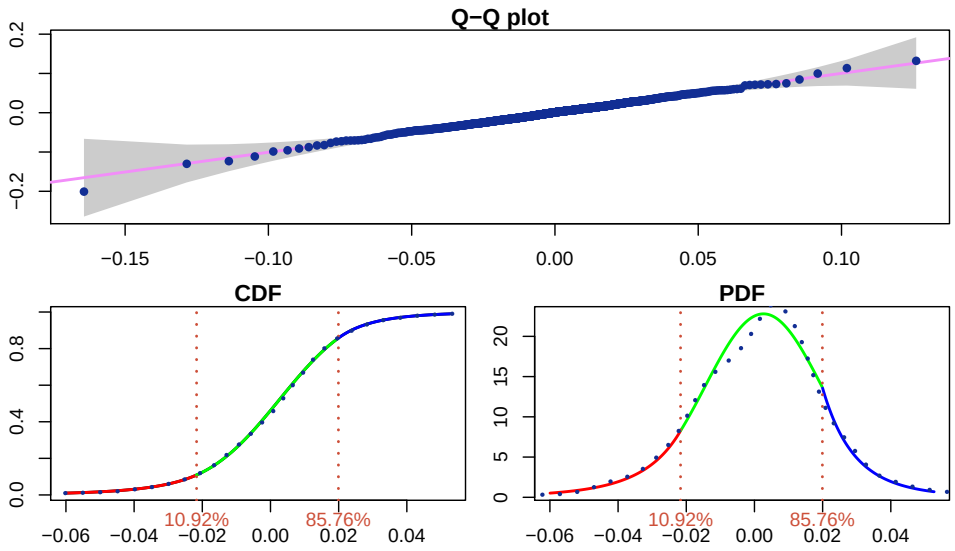


Figure 5: Fitted GNG model.

The log-likelihood increased to 6384.6 with the average of 2.4978 per data-point. In this model, the generalized Pareto distribution explains the first 13% from the left tail and the last 14% from the right tail. In addition, since GPD generalizes the Pareto distribution, the higher likelihood is a reasonable result.

Risk Measures: Package **mistr** provides a function, `risk()`, which can be used for rapid calculations of point estimates of prescribed quantiles, expected shortfalls and expectiles. As an input parameter this function needs the output of the function `PNP_fit()` or `GNG_fit()` from the same package. As an example we will illustrate these functions on our fitted objects.

```
> risk(PNP_model, alpha = c(0.05, 0.1, 0.12))
```

```
  level        VaR         ES        Exp
1  0.05 0.03215013 0.05701704 0.02723856
2  0.10 0.02328550 0.04207027 0.01882883
3  0.12 0.02102324 0.03874562 0.01679579
```

```
> risk(GNG_model, alpha = c(0.02,0.03, 0.05), plot = TRUE, size = 0.7)
```

```
  level        VaR         ES        Exp
1  0.02 0.04735437 0.06777767 0.03545433
2  0.03 0.04058584 0.05975390 0.03049295
3  0.05 0.03264750 0.05034339 0.02466034
```

